



United International University
Department of Computer Science and Engineering
 Mid Examination Trimester- Fall – 2024
 Course: Math -2205 (Probability and Statistics)
 Total marks - 30; Duration – 1 hour 30 Minutes

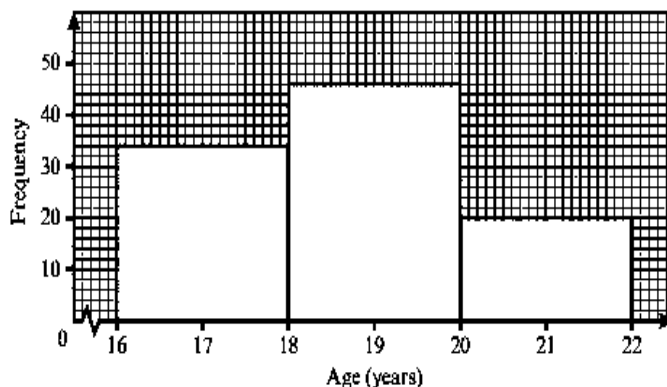
- Q1** A health researcher wants to study the relationship between the number of hours spent exercising per week and the cholesterol level of adults. A random sample of 8 individuals provides the following data:

Hours Exercise (x)	1	2	3	4	5	6	7	8
Cholesterol level (y)	2.4	2.2	2	1.95	1.90	1.80	1.75	1.70

- Find the Correlation Coefficient (r) and interpret the result. [4]
- Calculate r^2 and interpret it in the context of the study. [1]
- Determine the equation of the regression line in the form $y = a + bx$, where a and b are constant to be found. Explain the meaning of the slope b in the context of this scenario. [4]
- Use the regression equation to predict the cholesterol level for someone who exercises 10 hours per week. [1]

Q2

- The histogram represents the frequency distribution of individuals within different age groups, ranging from 16 years to 22 years. The x-axis represents the age groups, while the y-axis represents the frequency of individuals in each group.

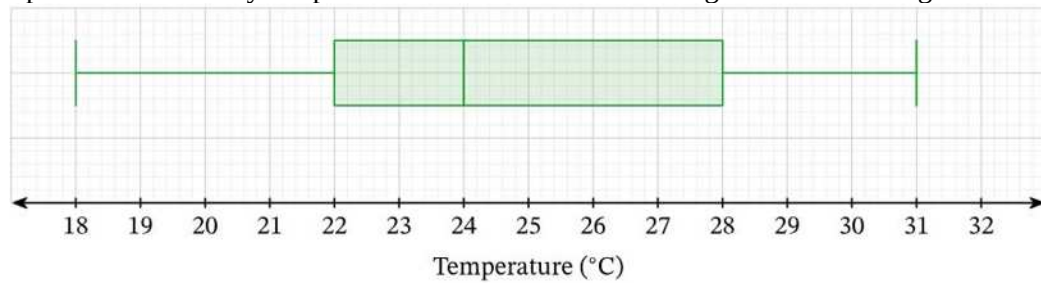


- What is the total frequency of the data represented in the histogram [1]
- What is the total number of individuals aged 19 and 21 combined? . [1]
- Calculate the percentage of individuals in the age group 16–17 [1]
- Give one disadvantage of using histogram. [1]

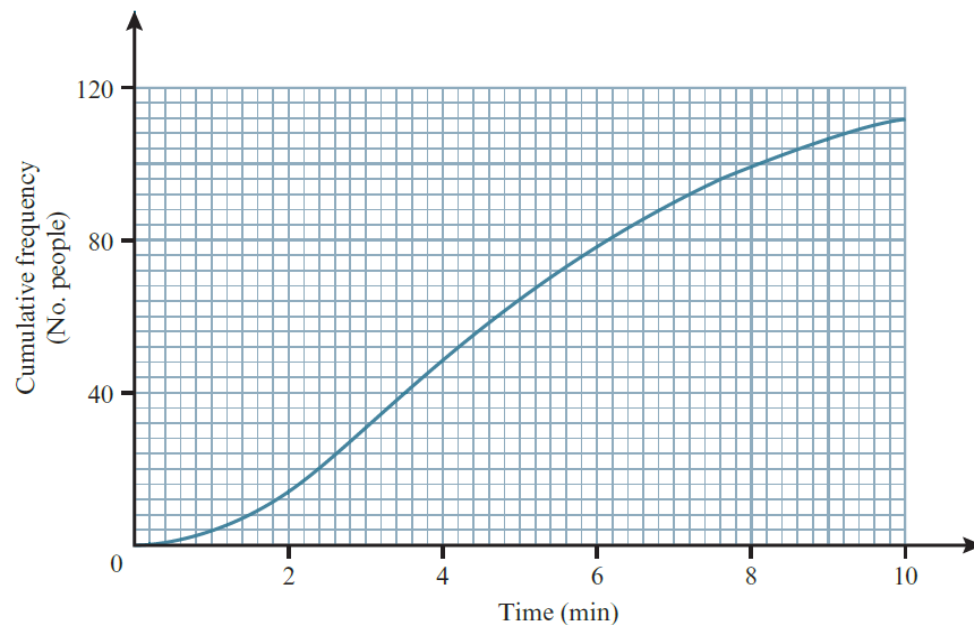
- A teacher recorded the quiz marks of 8 students as:
 11 13 15 15 17 18 19 20.
 He later realized that there was a typing error, so he changed the marks of 11 to 1. Discuss the effect on mode, mean and median of the students' marks. [2]
- The following table shows the number of **pet** owned by each of 20 families. Find the standard deviation and coefficient of variation (CV) . [4]

No of Pets	0	1	2	3	4	5
Number of Fm.	3	4	2	6	2	3

The boxplot shows the daily temperatures at a seaside resort during the month of August.



- (a) Find the range and inter quartile range. Also comment about the nature of the distribution. [3]
 (b) The following graph illustrates the times taken by 112 people to complete a puzzle.



- (i) Estimate what percentage of people took more than 8 minutes to complete. [3]
 (ii) Find, to an approximate degree of accuracy, an estimate of 70th Percentile. [1]

(c)

Girls (12)		Boys (13)	
4 1	8	2	
8 6 6	8	5 9	
3 2 1 0	9	0 1 3 3 4 4	
8 7 7	9	5 6 6 9	

Key: 1 | 8 | 2
 represents
 81% for a
 girl and 82%
 for a boy

This back-to-back stem and leaf diagram shows the percentage scores of the 25 students who were the top performers in the examination.

Identify any possible outliers. [3]

Formulae: